GeoGenBench: Symbolically Verified Generation of Geometry Diagrams

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Abstract

Geometry benchmarks ask a model to read a diagram. None asks it to draw one. GeoGenBench does. It has 801 prompts, each asking for a plane-geometry construction and each paired with the properties a correct drawing must have. The model writes TikZ. We compile it, recover the coordinates, and check every property exactly. No human grades and no vision model judges. We test six frontier models. Asking for a typed intermediate representation instead of raw TikZ lifts every model, by 2 to 25 points, most for the weakest. Under that scaffold the three best models tie within one point at a 7× cost spread. A human check of 200 outputs finds the verifier lenient in one direction only: it never fails a correct drawing. We release the dataset, the templates, the verifier, and the leaderboard.

1 Introduction

Language models now write worksheets and slides, and in school the picture they most often need to draw is a geometry diagram. Every geometry benchmark tests the other direction: here is a diagram, answer a question. None tests whether a model can draw a correct diagram from a description.

Take the prompt *"Draw an acute triangle EFG with angles* $60°$ *and* $70°$ *at E and F. Construct the altitude from G meeting EF at H."* A correct drawing has those two angles, H on the segment EF and not beyond it, a right angle at H, and no obtuse angle. Each of those is a check we can run on the coordinates of the model's drawing. In our pilot, one flagship model put H beyond the segment. Another drew an obtuse triangle. The verifier names the check that failed.

Two ideas make this work. The prompt and its checks come from the same template, so the checks can never ask for something the prompt did not. And the drawing is graded from its coordinates, so "right angle" is a calculation, not an opinion about how the picture looks.

Prior work. Geometry3K [6], GeoQA [3], UniGeo [4], MathVista [7], GeoEval [10], and GeoBench [1] test diagram reading. T2I-CompBench++ [5] and GenExam [9] grade generated images with a model as judge, which can say whether a picture looks like a right triangle but not whether it has a right angle. AlphaGeometry [8] and the Lean corpora machine-check proofs. GeoGenBench machine-checks drawings.

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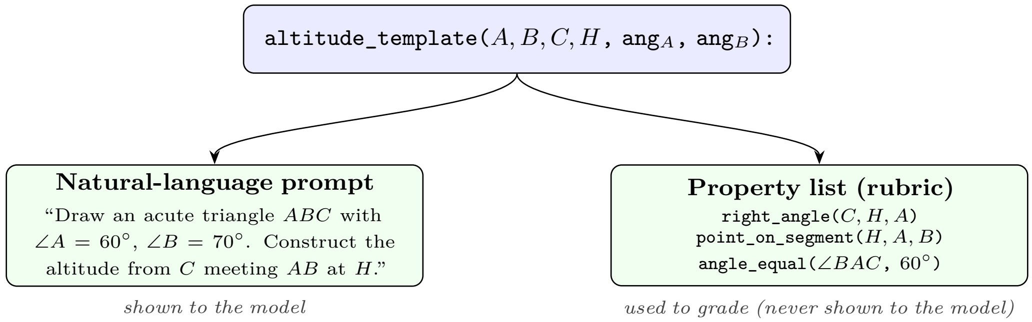

Figure 1: Procedural co-generation: one Python template emits both the prompt and the rubric, guaranteeing they stay mutually consistent.

2 The benchmark

Co-generation. One Python template writes both the prompt and its checks (Figure 1). The altitude template returns the sentence the model sees and the list `[right_angle(`A`,`H`,`B`), point_on_segment(`H`,`B`,`C`)]` the grader uses. The two cannot drift apart. A reviewer worried about contamination can re-roll the template with a fresh seed.

Symbolic verification. The model emits TikZ. We compile it twice: once to an image, once to SymPy geometry with coordinates. Fifteen checks such as `right_angle`, `midpoint`, `parallel`, and `point_on_segment` run on those coordinates in microseconds, with a small numeric tolerance. Two weaker checks on labels and marks count only toward a loose pass. Template authors say what must hold, not how to test it.

Scoring. The model sees only the prompt. An output passes if every check passes. It gets a soft pass if nothing fails but a check was skipped. Otherwise it fails, or it never rendered.

Composition. 600 of the 801 prompts come from 30 templates, 200 per tier: T1 is a single shape, T2 adds one construction, T3 combines several. The other 201 were extracted by a language model from a state K–12 curriculum [2]. We report the two sources separately and treat the templated 600 as the anchor.

3 Verifier reliability

Completeness. A check list is complete if every drawing that passes it is the intended construction, up to position and scale. Of the 30 templates, 18 are that tight. The other 12 pin the named relations but leave free parameters, as a teacher's instructions usually do. We also tried to fool each check with TikZ that passes it while contradicting the prompt. Five checks can be fooled alone. Templates pair checks, so we estimate under 5% of passes are exploitable.

Hardening. Three verifier fixes promoted 4.6% of pilot outputs from soft pass to pass and demoted none. They also halved the apparent gap between the two strategies. Part of that gap had been the verifier's, not the models'. Every number below is post-fix.

Human check. Two raters, a PhD mathematician and an engineer who has taught mathematics, judged 200 outputs blind to the verifier's verdict. Verifier and raters agree moderately ($\kappa = 0.455$), below the 0.70 we had pre-registered, while the raters agree well with each other. The raters overturned 18% of the verifier's passes and none of its fails. We return to this in Section 5.

Table 1: Headline leaderboard (600 scenarios × 3 repeats, post-hardening). Strict = `pass`; loose adds `soft_pass`. * marginally partial coverage; † blocked at the API layer (Section 4).

Model	Strategy	N	Strict	Loose	$/scen.	Latency
GPT-5.5	structured	2,304*	**94.1%**	94.1%	$0.075	33.6 s
Claude Sonnet 4.6	structured	2,861	93.4%	93.4%	$0.088	19.0 s
Claude Opus 4.7	structured	2,292*	93.3%	93.3%	$0.532	23.9 s
GPT-5.5	raw_code	1,800	91.9%	95.3%	$0.058	14.8 s
Gemini 2.5 Pro	structured	1,140*	83.1%	83.1%	$0.050	75.7 s
Claude Sonnet 4.6	raw_code	1,800	81.2%	95.3%	$0.044	16.4 s
Claude Haiku 4.5	structured	2,232	75.3%	76.8%	$0.032	8.0 s
Claude Opus 4.7	raw_code	1,800	71.6%	95.3%	$0.262	13.8 s
Claude Haiku 4.5	raw_code	2,403	70.5%	76.0%	$0.019	9.5 s
Gemini 2.5 Pro	raw_code	1,800	58.4%	89.9%	$0.049	32.9 s
Gemini 2.5 Flash	raw_code	2,403	35.9%	42.2%	$0.012	22.1 s
Gemini 2.5 Flash	structured	0†	—	—	—	—

4 Results

Setup. Six models from three vendors at a 20× price spread: Claude Opus 4.7, Claude Sonnet 4.6, Claude Haiku 4.5, GPT-5.5, Gemini 2.5 Pro, and Gemini 2.5 Flash. Two strategies. raw_code hands the model the prompt and a TikZ tutorial and takes what it writes. structured asks for typed JSON describing points, segments, circles, and checks, which our code turns into SymPy and then TikZ. The checks run before rendering, so the model gets up to three retries. Every model-by-strategy cell runs on the 600 templated prompts three times (Table 1).

The scaffold lifts every model. structured helps each model: GPT-5.5 by 2 points, Claude Haiku 4.5 by 5, Claude Sonnet 4.6 by 12, Claude Opus 4.7 by 22, and Gemini 2.5 Pro by 25. The weaker a model is at raw TikZ, the more it gains.

The frontier converges. With the scaffold, GPT-5.5, Claude Sonnet 4.6, and Claude Opus 4.7 finish at 94%, 93%, and 93%, within one point of each other. Claude Opus 4.7 costs seven times what GPT-5.5 does. Without the scaffold the spread runs from 92% down to 36%, and Claude Opus 4.7 is the most dominated cell, 20 points behind GPT-5.5 at four times the cost.

Difficulty tiers separate the strategies, then the models. T1 is nearly solved by every Anthropic and OpenAI model under the scaffold. T2 is where the scaffold first matters: it lifts Claude Opus 4.7 by 43 points and Gemini 2.5 Pro by 54. T3 is where the frontier models separate: GPT-5.5 93%, Claude Sonnet 4.6 87%, Claude Opus 4.7 82%.

Wrong drawings, not broken code. Most failures are clean, rendered diagrams with the wrong geometry. The verifier catches them; a human glancing at the picture often would not.

Portability. structured needs schema-constrained decoding. Our schema compiles to thousands of decoder states, and one vendor's API refuses it before the model writes a token. That is a gap in the API, not the model, and a warning for any benchmark built on a large schema.

5 Discussion

Exact grading is practical. Every drawing in Table 1 was graded by a program, in microseconds, with the same verdict every time. A model judge would have cost roughly a thousand times more per drawing and could not have told a right angle from one that looks right.

The scaffold matters more than the model. The largest single effect in the study is not which model you buy but whether you ask it for a typed representation before it draws. That representation lets the model check its own work before rendering. The most expensive model gains the most from it and, once it has it, no longer earns its price.

The verifier errs in one direction only. The human study found the verifier lenient, never harsh. Every drawing it fails is wrong, and about one pass in six is not as right as it looks, almost always on T2, where the checks confirm the construction but not the numbers. That lenience applies to every model on the same prompts, so the rankings hold even where the absolute rates are a little high. Numeric checks for those templates are the planned fix.

Limitations. GeoGenBench tests whether a model can turn a sentence into a consistent plane drawing. It does not test proofs, solids, or teaching. A third of the templates accept any drawing that meets the stated constraints, not one canonical drawing. The curriculum prompts are harder for every model than the templated ones, and the headline numbers are templated numbers.

Takeaway. A geometry drawing can be graded exactly, from its coordinates, with no judge. Graded that way, the scaffold matters more than the model. We release everything needed to run it.

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